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**Anurag(Anu) Karuparti** · 3rd+

Agentic AI Strategist @Microsoft (25k+) | Author - Generative ...

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I have spent the last year helping Enterprises move from "IMPRESSIVE DEMOS" to "RELIABLE AI AGENTS".

The pattern is always the same:

Teams nail the LLM integration and think the hard part is done, then realize they have built 20% of what production actually requires.

Here is why each building block matters:

Reasoning Engine (LLM): Just the Beginning

- Interprets intent and generates responses
- Without surrounding infrastructure, it is just expensive autocomplete
- Real engineering starts when you ask: "How does this agent make decisions it can defend?"

Context Assembly: Your Competitive Moat

- Where RAG, memory stores, and knowledge retrieval converge
- Identical LLMs produce vastly different results based purely on context quality
- Prompt engineering does not matter if you are feeding the model irrelevant information

Planning Layer: What to Do Next

- Breaks goals into steps and decides actions before acting
- Separates thinking from doing
- Poor planning = agents that thrash or make circular progress

Guardrails & Policy Engine: Non-Negotiable

- Defines what APIs the agent can call, what data it can access
- Determines which decisions require human approval
- One misconfigured tool call can cascade into serious business impact

Memory Store: Enables Continuity

- Short-term state + long-term memory across interactions
- Without it, every conversation starts from zero
- Context window isn't memory it's just scratchpad

Validation & Feedback Loop: How Agents Improve

- Logging isn't learning
- Capture user corrections, edge cases, quality signals
- Best teams treat every interaction as potential training data

Observability: Makes the Invisible Visible

- When your agent fails, can you trace exactly why?
- Which context was retrieved? What reasoning path? What was the token cost?
- If you can not answer in under 60 seconds, debugging will kill velocity

- Cost & Performance Controls: POC vs Product
- Intelligent model routing, caching, token optimization are not premature they are survival
 - Monthly bills can drop 70% with zero accuracy loss through smarter routing

What most teams miss:
They build top-down (UI → LLM → tools)
when they should build bottom-up (infrastructure → observability → guardrails → reasoning).

These 11 building blocks are not theoretical. They are what every production agent eventually requires either through intentional design or painful iteration.

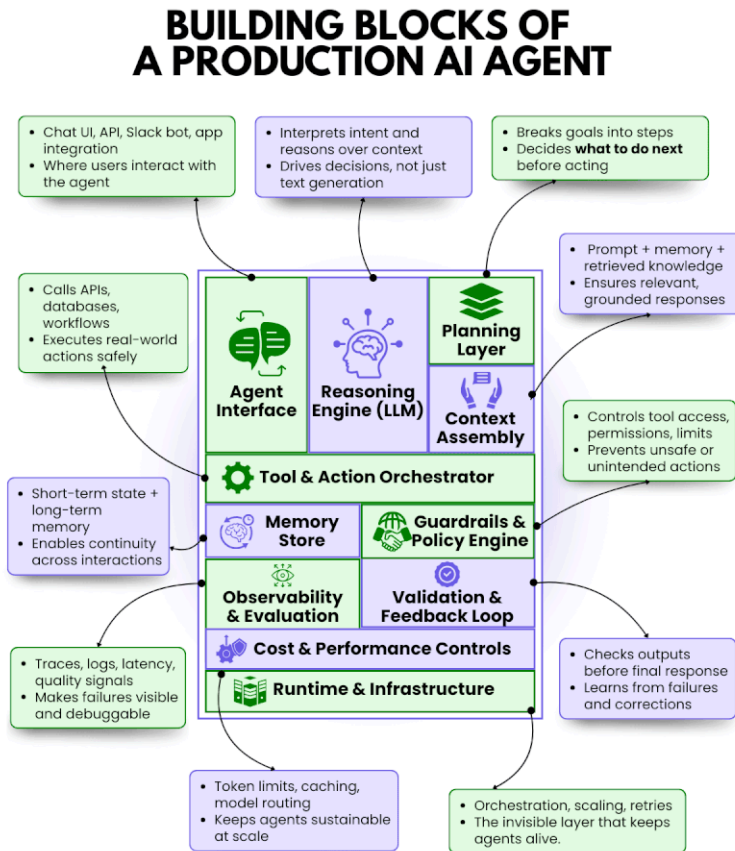
Which block are you currently underinvesting in?

- 🌱 Repost this to help your network get started
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PS: If you found this valuable, join my weekly newsletter where I document the real-world journey of AI transformation.

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Helping leaders architect AI agents at scale

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Jaswindder Kummar  · 2nd

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Director - Cloud Engineering | I design and optimize secure, scalable, and h...

Great breakdown [Anurag\(Anu\) Karuparti](#) These building blocks are crucial for turning impressive demos into reliable, production-ready AI agents.

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Anurag(Anu) Karuparti  Author

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Agentic AI Strategist @Microsoft (25k+) | Author - Generative AI for C...

Absolutely [Jaswindder Kummar](#) these layers turn flashy demos into systems that consistently perform under real-world conditions.

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Jegan Selvaraj  · 3rd+

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CEO @ Entrans Inc, Infisign Inc & Thunai AI | Enterprise AI | Agentic AI | MC...

The building blocks of production AI agents are easy to overlook,

Context assembly and guardrails often determine whether agents succeed or fail. [Anurag\(Anu\) Karuparti](#)

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Anurag(Anu) Karuparti  Author

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Spot on [Jegan Selvaraj](#) context and guardrails make or break agents in production.

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Sriram Yedida-Gold Medalist  · 3rd+

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PhD |AI & Data Science Educator | RAG | LLM |Agents| GenAI | Agentic AI | ...

Exactly — teams think "LLM integrated = done." The real work is in context, planning, and guardrails. Execution > demo every time.

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[Anurag\(Anu\) Karuparti](#) Exactly. Once you're in production, the model is the smallest decision. Everything around it determines reliability, cost, and trust.

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Prashant Varshney  · 3rd+

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Senior Software Engineer @ Intuit | Building Tech that Power Millions | Syste...

Wow [Anurag\(Anu\)](#). this is a great breakdown of the key building blocks for production AI agents. Most teams definitely miss some of these crucial steps!

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Spot on [Prashant Varshney](#) missing even one layer can turn a promising AI agent into an unstable system.

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Sivasankar Natarajan  · 3rd+

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Technical Director | GenAI Practitioner | Azure Cloud Architect | Data & Anal...

This reflects real field experience [Anurag\(Anu\) Karuparti](#) - Clear and respectful of the unglamorous work that makes agents dependable.

https://www.linkedin.com/posts/anuragsirish_genai-aiagents-activity-7421907388548468736-etJ9/?utm_medium=ios_app&rcm=ACoAAAmGi4BLk4EzkNCFtdNk...

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Happy it resonated! [Sivasankar Natarajan](#)

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Sonal Kurade  · 3rd+

Co-founder | DevOps Consultant | Cloud Architecture & Automation Special...

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This is a fantastic breakdown of what it really takes to build production-ready AI agents

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Glad this resonates! [Sonal Kurade](#)

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Krupal Chaudhary  · 3rd+

Founder @Demaze Tech | AI Product Builder | Empowering SMBs with AI-dri...

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This is exactly what teams discover the hard way [Anurag\(Anu\) Karuparti](#)
The demo proves intelligence but production demands accountability, traceability and control

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Right [Krupal Chaudhary](#) production thrives on governance, traceability, and control, not just demos.

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Keith R. Worfolk - MBA, MCIS, CCIO, CISSP, CCISO, C...  · 3r...

Head of Artificial Intelligence | Chief Technology Officer | CIO | Chief AI Offi...

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Completely agree, most teams stop at the LLM and underestimate everything that actually makes agents reliable.

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Yes [Keith](#) reliability comes from the systems and guardrails surrounding the model, not the model itself.

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Michael Lee  · 3rd+

CRO | Data & AI | Scaling \$1M-\$100M B2B Companies With AI | Turning Lea...

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Most teams confuse intelligence with reliability. Production agents succeed when decisions are explainable and reversible.

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Exactly [Michael Lee](#) reliability comes from clear, controlled decisions, not just apparent intelligence.

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Prem N.  · 3rd+

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Excellent breakdown, [Anurag\(Anu\)](#) - The focus on foundational infrastructure over just LLMs is key to AI agents success in production.

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https://www.linkedin.com/posts/anuragsirish_genai-aiagents-activity-7421907388548468736-etJ9/?utm_medium=ios_app&rcm=ACoAAAAMGi4BLk4EzkNCFtdNk... 4/5

Absolutely **Prem N.** strong infrastructure is the real foundation for production-ready AI agents.

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